

House Price Prediction with Multiple Linear Regression

Machine Learning Assignment 2

Objective: build a multiple linear regression model that predicts House_Price_Million_RWF from house characteristics in the supplied Rwanda real-estate dataset.

1. Data Exploration and Cleaning

The raw dataset contains 125 rows and 9 columns. There were 5 duplicate records and missing values in several columns. House_ID was treated as an identifier rather than a predictor. Exact duplicates were removed, non-positive/impossible observations were checked, rows with a missing target were removed, and IQR-based outlier rows were removed. The final cleaned dataset contains 91 rows.

Problem found	Where	Action	Reason
Duplicate records	Dataset	Remove exact duplicates	Avoid counting the same house sale more than once.
Missing predictor values	Numeric/categorical predictors	Drop from pipeline	Retain observations while avoiding manual arbitrary replacement.
Missing target	House_Price_Million_RWF	Remove row	A supervised model cannot train without the target.
IQR outliers	Numeric predictors and target	Remove outlier rows	Reduce influence of extreme observations identified by the IQR rule.
House_ID	Identifier column	Exclude from model	It is an ID, not a house characteristic.

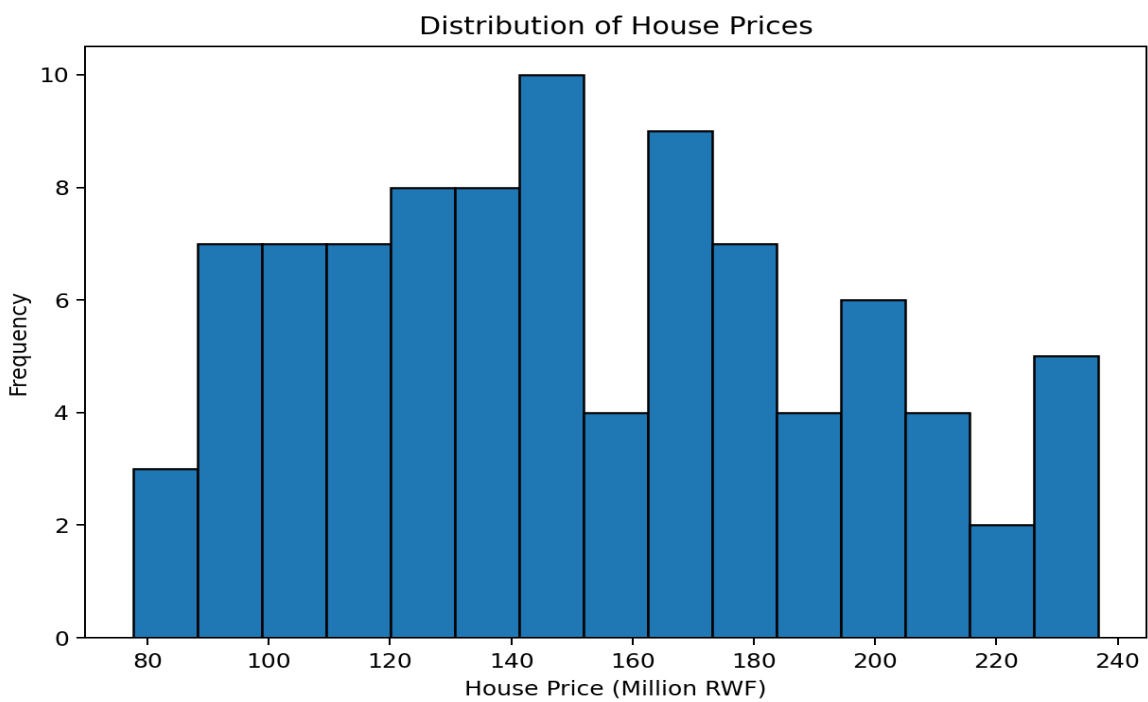


Figure 1. Distribution of house prices.

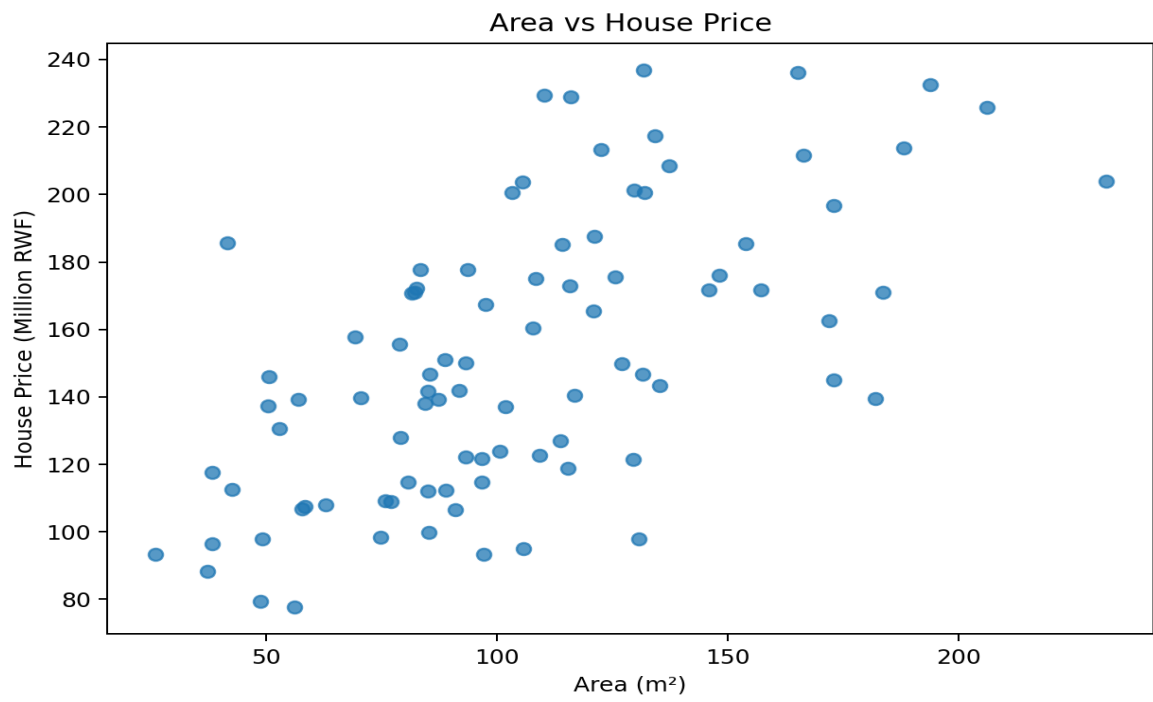


Figure 2. Relationship between area and house price.

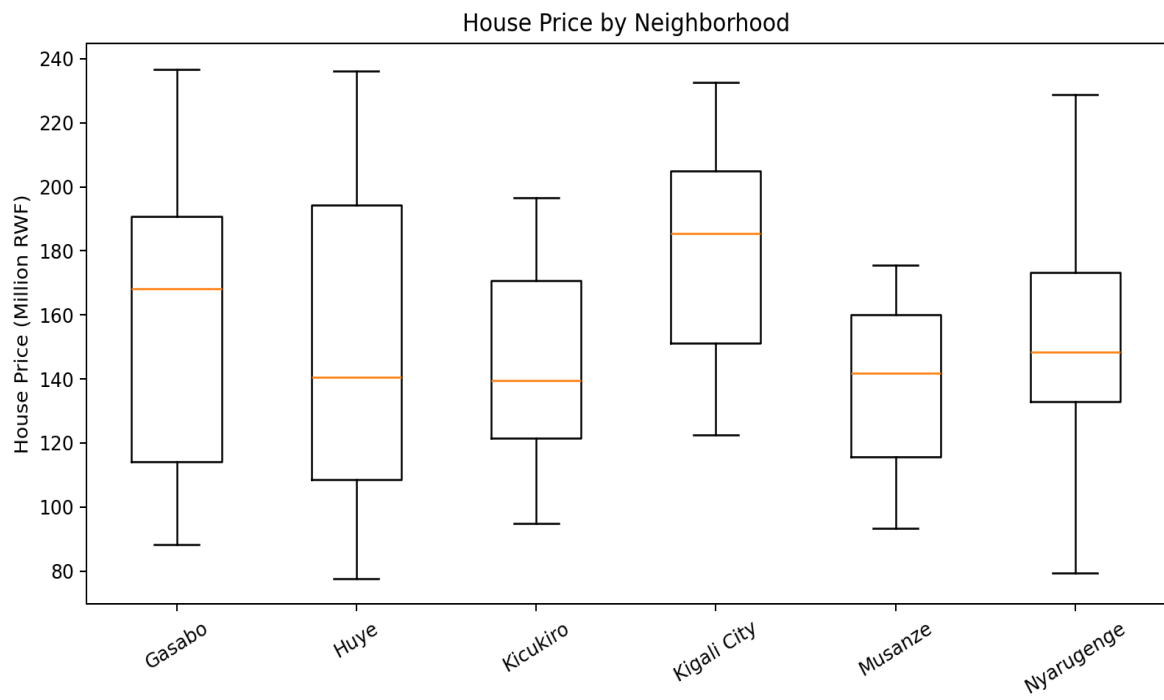


Figure 3. House price by neighborhood.

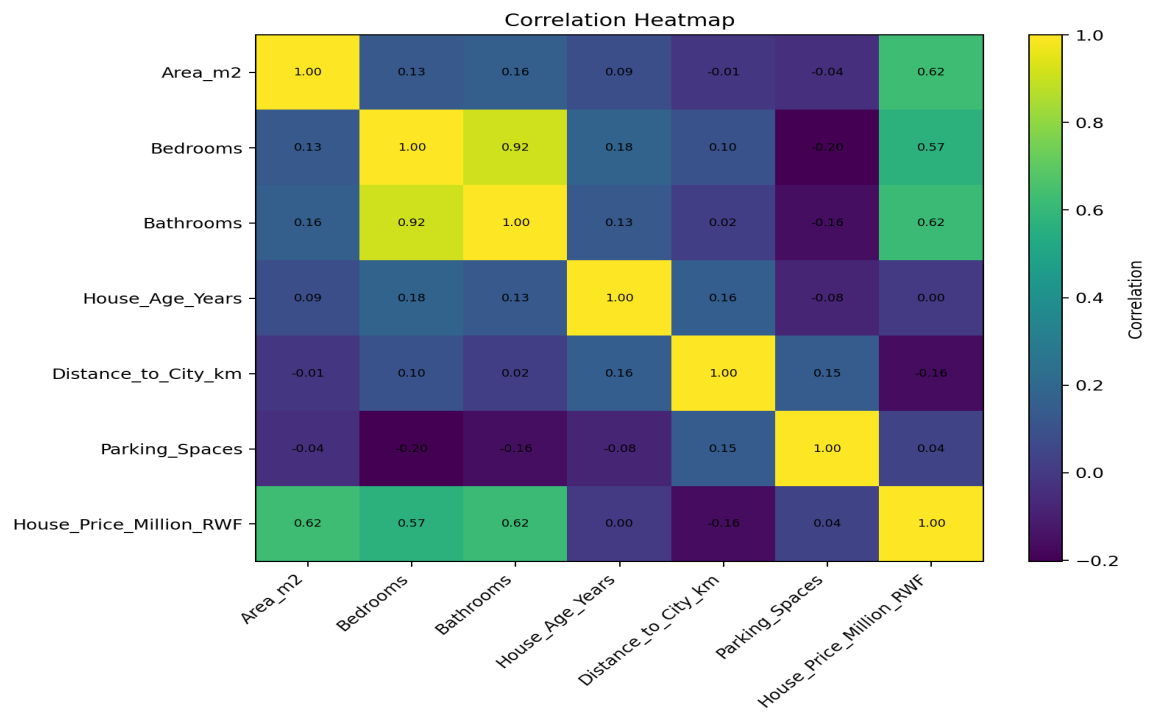


Figure 4. Correlation heatmap.

2. Multiple Linear Regression Model

The predictors are Area_m2, Bedrooms, Bathrooms, House_Age_Years, Distance_to_City_km, Parking_Spaces, and Neighborhood. Neighborhood is categorical and is one-hot encoded with one category dropped to avoid the dummy-variable trap. The data is split into 80% training and 20% testing using random_state=42. The complete preprocessing and regression model are stored in one Pipeline.

Intercept: 48.3256

Variable	Coefficient
num__Area_m2	0.5372
num__Bedrooms	9.2098
num__Bathrooms	12.2959
num__House_Age_Years	-1.0562
num__Distance_to_City_km	-1.5937
num__Parking_Spaces	5.9376
cat__Neighborhood_Huye	-15.2642
cat__Neighborhood_Kicukiro	-21.4506
cat__Neighborhood_Kigali City	0.4502
cat__Neighborhood_Musanze	-20.9918
cat__Neighborhood_Nyarugenge	-5.4101

3. Model Evaluation

Dataset	R ²	Adjusted R ²	MAE	RMSE
Training	0.7666	0.7238	15.8410	19.6091
Testing	0.7759	0.4237	16.3259	18.8698

On the test set, the model produced $R^2 = 0.7759$, MAE = 16.33 million RWF, and RMSE = 18.87 million RWF.

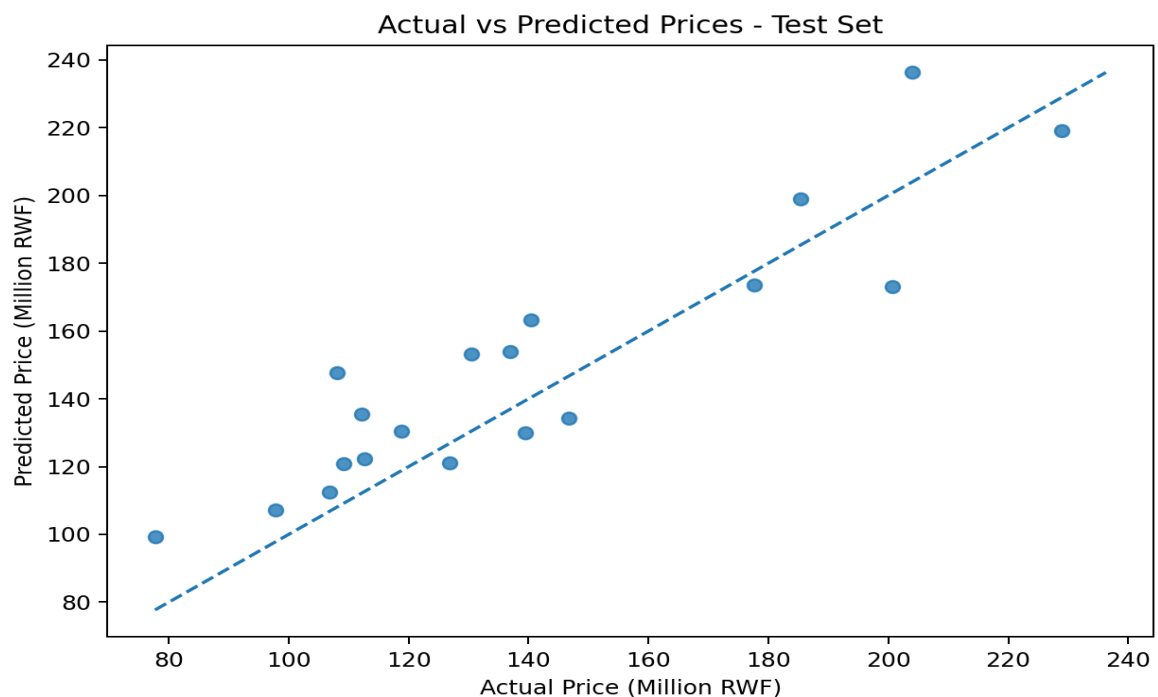


Figure 5. Actual versus predicted test prices.

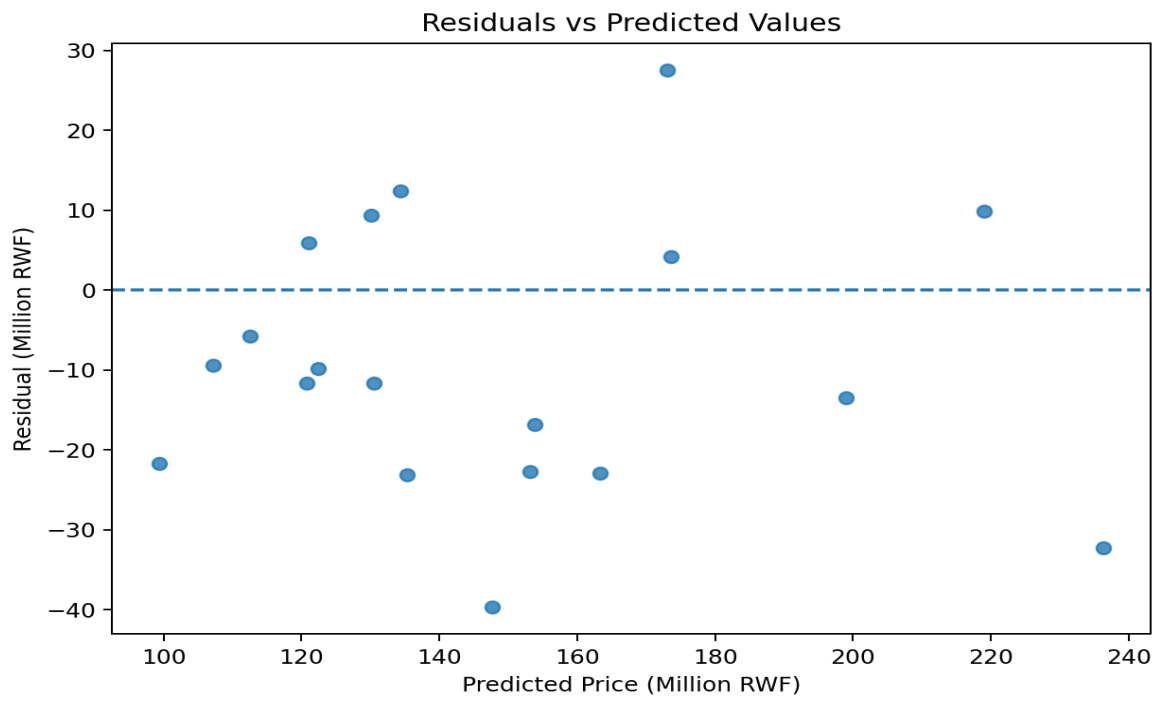


Figure 6. Residuals versus predicted values.

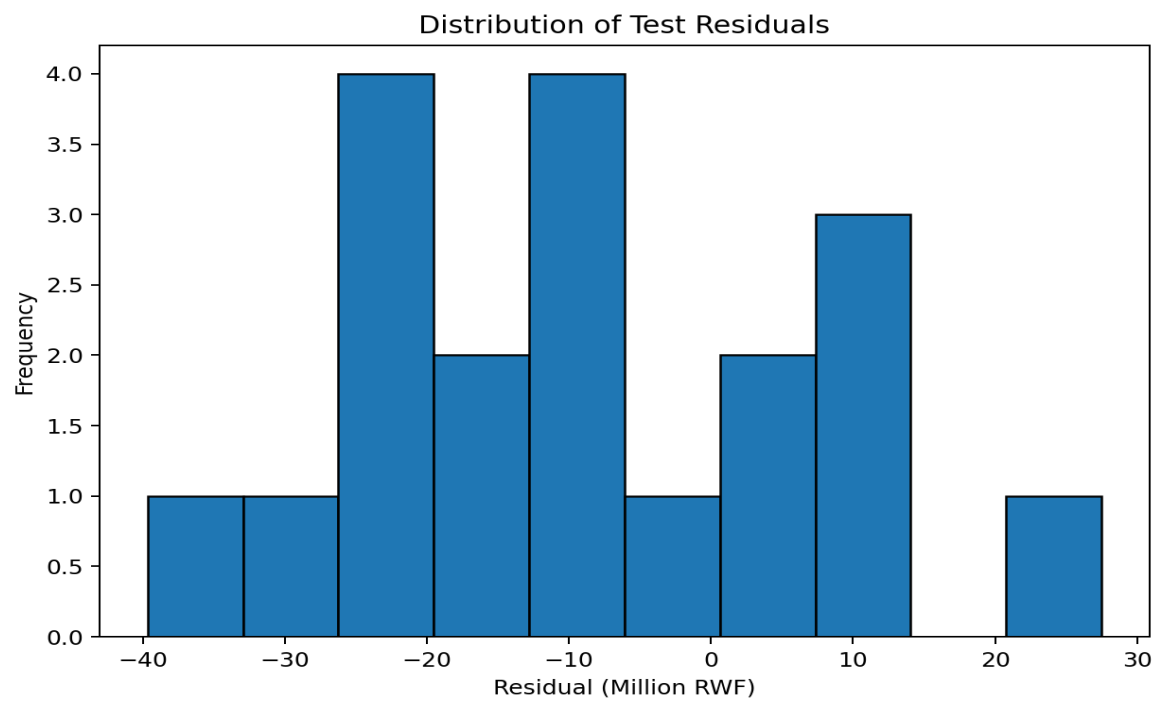


Figure 7. Distribution of test residuals.

4. Interpretation

Holding the other predictors constant, the Area_m2 coefficient is 0.5372, meaning an additional square metre is associated with an estimated 0.5372 million RWF change in predicted price. The Bedrooms coefficient is 9.2098, and the Bathrooms coefficient is 12.2959 million RWF per additional unit, holding other variables constant. House age has a coefficient of -1.0562 million RWF per additional year.

The correlation matrix is used as a simple multicollinearity diagnostic. Strong relationships among predictors should be monitored because correlated predictors can make individual coefficient estimates less stable. The residual plots are used to assess whether residual spread is approximately constant and whether there are strong patterns that would contradict the linear model.

Limitations: (1) the dataset is relatively small, so estimates may change with another sample; (2) removing IQR outliers can discard legitimate high-value or unusual houses and therefore may limit generalization to extreme properties.

5. Model Saving and New Prediction

The trained pipeline was saved as house_price_model.sav and loaded again. Prediction equality after reloading: True. For the example house (150 m², 3 bedrooms, 2 bathrooms, 5 years old, 4 km from the city centre, 1 parking space, Gasabo), the loaded model predicts approximately 175.40 million RWF.

6. Streamlit Application and Deployment

The supplied app.py loads the saved pipeline using @st.cache_resource, provides input widgets for all seven model features, validates key numeric ranges, and displays the prediction in million RWF. Run it locally with: streamlit run app.py. The GitHub and Streamlit Community Cloud steps must be completed using the student's own GitHub account; screenshots of the running app should be added before final submission.

LIVE APPLICATION

The completed house price prediction model is deployed online as a Streamlit application. Users can enter the seven property characteristics and obtain an estimated house price directly through the web application.

Live Streamlit Application

<https://01a0fc20-d767-8a23-da82-b8de5be90964.share.connect.posit.cloud/>

The application is hosted through Posit Connect and is available online at the link above.